

HORN: Directional Outlier Propagation in Hybrid Reasoners

Anonymous authors

Paper under double-blind review

Abstract

This is a living paper shell for a preregistered Mac-first experiment. HORN tests whether attention-to-SSM and SSM-to-attention boundaries differ in outlier magnitude and quantization-noise sensitivity. No positive result or native systems claim is made yet.

1 Current Gate

The branch must pass activation-magnitude, quantization-noise propagation, and cross-model/control gates before GPU validation.

2 Mac Artifact Status

A deterministic synthetic H1 packet validates the directional-boundary readout. It reports SSM-to-attention over attention-to-SSM max ratio 3.775 and kurtosis ratio 7.139, producing `SYNTHETIC_PASS_REAL_BOUNDARY_DUMPS_NEXT`. This is synthetic-only: it is not model evidence, not a quantization-noise result, and not GPU evidence.

3 Next Evidence

The next admissible row is a real boundary activation dump from a hybrid model. HORN remains a control branch unless real H1/H2/H3 gates show consistent directional asymmetry that pure-attention and pure-Mamba controls do not share.